Exercise 1

Q.4 Given that

$Cov(X_t, X_{t-k}) = \gamma_k$ is free of t,

but $E(X_t) = 3t$ is dependent on t.

(a) $\{X_t\}$ is not stationary since $E(X_t) = 3t$ which is dependent on t.

(b) $Z_t = 7 - 3t + X_t$

$E(Z_t) = E(7 - 3t + X_t) = 7 - 3t + E(X_t)$

$= 7 - 3t + 3t = 7$

$Cov(X_t, X_{t-k}) = E(X_t X_{t-k}) - E(X_t)E(X_{t-k})$

$= E(X_t X_{t-k}) - 3t[3(t-k)] = \gamma_k$

$\therefore E(X_t X_{t-k}) = \gamma_k + 9t(t-k)$

If $k = 0$, then $E(X_t^2) = \gamma_0 + 9t^2$.

Hence $V(Z_t) = V(X_t) = E(X_t^2) - [E(X_t)]^2$

$= \gamma_0 + 9t^2 - (3t)^2$

$= \gamma_0$

$\therefore$ Z_t is stationary.

Q.5 $Y_t = A + X_t$, A and X_t are independent and $\{X_t\}$ is stationary.

$$E(Y_t) = E(A + X_t) = E(A) + E(X_t) = \mu_A + \mu_X \quad (\text{is free of } t)$$

$$Cov(Y_t, Y_{t+k}) = Cov(A + X_t, A + X_{t+k})$$

$$= Cov(A, A) + Cov(A, X_{t+k}) + Cov(X_t, A) + Cov(X_t, X_{t+k})$$

$$= V(A) + 0 + 0 + Cov(X_t, X_{t+k}) = V(A) + Cov(X_t, X_{t+k})$$

which is free of t since $\{X_t\}$ is stationary (i.e., $E(X_t)$, $V(X_t)$ and $Cov(X_t, X_{t-k})$ all do not depend on t)

$\therefore$, Y_t is stationary.

Q.6 $Y_1 = \theta_0 + e_1$, $t > 1$ (the given model)

$$Y_t = \theta_0 + Y_{t-1} + e_t \quad (1)$$
$$Y_{t-1} = \theta_0 + Y_{t-2} + e_{t-1} \quad (2)$$
$$Y_{t-2} = \theta_0 + Y_{t-3} + e_{t-2} \quad (3)$$
$$\vdots$$
$$Y_1 = \theta_0 + Y_0 + e_1 \quad (t)$$

$$Y_t = \theta_0 + \theta_0 + Y_{t-2} + e_{t-1} + e_t$$
$$= 2\theta_0 + \theta_0 + Y_{t-3} + e_{t-2} + e_{t-1} + e_t$$
$$= 3\theta_0 + Y_{t-3} + \sum_{i=0}^{2} e_{t-i}$$

$Y_0 = 0$ $\because$ the given model.

Substitute (2) – (t) into (1)

$$\therefore Y_t = (t-1)\theta_0 + Y_1 + e_2 + e_3 + \cdots + e_t$$
$$= (t-1)\theta_0 + \theta_0 + e_1 + e_2 + e_3 + \cdots + e_t$$
$$= t\theta_0 + \sum_{i=1}^{t} e_i$$

$$E(Y_t) = E\left(t\theta_0 + \sum_{i=1}^{t} e_i\right) = t\theta_0 + 0 = t\theta_0 \quad (\text{depends on } t)$$

Q.6 (cont'd)

$$Cov(Y_t, Y_{t-k}) = Cov\left(t\theta_0 + \sum_{i=1}^{t} e_i,\ (t-k)\theta_0 + \sum_{i=1}^{t-k} e_i\right)$$

$$= Cov\left(\sum_{i=1}^{t} e_i, \sum_{i=1}^{t-k} e_i\right) \quad (*)$$

$$= Cov\left(\sum_{i=1}^{t-k} e_i, \sum_{i=1}^{t-k} e_i\right) = \sum_{i=1}^{t-k} Cov(e_i, e_i)$$

$$= \sum_{i=1}^{t-k} V(e_i) = \sum_{i=1}^{t-k} \sigma_e^2$$

$$= (t-k)\,\sigma_e^2$$

Also the covariance between Y_t and Y_{t-k} is dependent on t. $\therefore$ Y_t is NOT stationary.

(*) notice that $Cov(e_t, e_{t-k+1}) = 0$ and onwards

